MIL-M-10958F

29 July 1976
SUPERSEDING
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MILITARY SPECIFICATION

MARGARINE, CANNED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a vegetable-type margarine for use by the Armed Forces as an item of general issue and as a component of operational rations (see 6.1).

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-C-29 - Canned Subsistence Items, Packaging and Packing Of

MILITARY

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-668 - Sanitary Standards for Food Plants

FSC 8945

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(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

US DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder
Food Additive Regulation, Sec. 121.1017 and .1056
Definitions and Standards of Identify for Margarine, Oleomargarine

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Officer, Washington, DC 20402.)

US DEPARTMENT OF AGRICULTURE

US Standards for Grades of Nonfat Dry Milk

General Specifications for Dairy Plants Approved for USDA Inspection and Grading Service

(Application for copies should be addressed to the US Department of Agriculture, Dairy Division, AMS, Washington, DC 20250.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the effect on date of invitaton for bids or request for proposal shall apply.

THE UNITED STATES PHARMACOPOEIAL CONVENTION, INC.

Pharmacopoeia of the United States

(Application for copies should be addressed to the Mack Publishing Company, 20th and Northampton Streets, Easton, PA 18042.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS

Official Methods of Analysis of the
Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, P.O. Box 540, Benjamin Franklin Station, Washington, DC 20044.)

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AMERICAN OIL CHEMISTS' SOCIETY

Official and Tentative Methods

(Application for copies should be addressed to the American Oil Chemists' Society, 508 South Sixth Street, Champaign, IL 61820.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

* 3.1 Material. The material shall comply with 3.1.1 and table I.

* 3.1.1 Milk. The raw milk or milk products used in the preparation of this product shall comply with all the quality requirements for No. 2 (or better) specified in the General Specifications for Dairy Plants Approved for USDA Inspection and Grading Service. Undergrade or probational milk shall not be used. The milk shall be drawn from cows in herds accredited as tuberculosis-free and certified brucellosis-free by the US Department of Agriculture, or herds that have passed an annual tuberculosis test and meet USDA requirements for an individually certified herd, or from cows in herds located in:

(1) A modified accredited tuberculosis area; and
(2) Either (a) A certified brucellosis area; or
(b) Modified certified brucellosis area; or
(3) An area in the process of being certified by the USDA in accordance with USDA Animal and Plant Health Inspection Service publication 91-1.

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* TABLE I. Components

Item	Requirement
1.	Cottonseed oil shall be of good quality, properly refined, partially hydrogenated and deodorized. One or more of the additives such as, citric acid, isopropyl citrate or stearyl citrate may be incorporated into the oil. In addition, the cottonseed oil shall have a fat stability (AOM) of not less than 65 hours and a peroxide value of not more than 1.0 milliequivalents per 1000 grams of oil at time of formulation of the margarine. 1/
2.	Water used shall meet drinking water standards approved by State or Federal regulatory agencies and shall have no undesirable flavors or odors.
3.	Salt shall be refined, crystalline, free-flowing, evaporated sodium chloride and of the type commonly used in margarine. Iodized salt shall not be used.
4.	Milk solids-not-fat shall be obtained from extra grade nonfat dry milk and at the time of use shall meet the requirements for extra grade as defined in U. S. Standards for Grades of Nonfat Dry Milk and shall be certified as salmonella negative.
5.	Other optional ingredients permitted in the Definitions and Standards of Identity for Margarine, Oleomargarine and limited to emulsifying and preservative agents (see table III), artificial flavoring substances in semblance of butter, color additives and vitamin A shall be used in the formulation.
6.	Vitamin D, when specified, shall be added in such quantity that the finished product complies with the analytical requirements of 3.3.1.

1/ Determined on the oil (tests run to completion and results reported to the nearest hour) not more than 96 hours prior to use in production of the margarine. Determination shall be made subsequent to addition of emulsifying and anti-oxidant materials as specified in item 5 of table I.

* 3.2 Processing. The product shall be prepared from the materials specified in 3.1.1 and table I, and in accordance with the formulation necessary to meet the analytical requirements of table III. In addition, both sanitary methods and equipment (see 3.6) and operating procedures reflecting good commercial practice shall be used.

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3.3 Finished product. The finished product shall comply with the requirements listed in table II.

TABLE II. Finished product

Item	Requirement
1.	The product shall have a uniform yellow color characteristic of colored commercial margarine.
2.	There shall be no foreign flavor such as rancid, tallowy or oxidized.
3.	The product shall have a firm compact body of good spreadability at 60° - 70°F (16 to 21°C), and shall not be crumbly, grainy, mealy, oily or visibly moist.
4.	The finished product shall have been processed in such a manner as to pass the cube test specified in 4.5.3 before release for shipment from the contractor's plant.

3.3.1 Analytical requirements. The analytical requirements shall be as listed in table III.

TABLE III. Analytical requirements

Item	Requirement
1.	Fat - not less than 80.0 percent.
2.	Salt - not less than 2.5 nor more than 3.0 percent.
3.	Vitamin A - not less than 15,000 I.U. units per pound (0.45 kg).
4.	Vitamin D - not less than 1500 I.U. units per pound (0.45 kg) (when applicable).
5.	Potassium sorbate - not less than 0.08 nor more than 0.10 percent. 1/
6.	Propyl gallate - not less than 45 nor more than 50 parts per million (ppm). 1/

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TABLE III. Analytical requirements (cont'd)

Item	Requirement
7.	Butylated hydroxyanisole - not less than 70 nor more than 75 ppm. 1/
8.	Butylated hydroxytoluene - not less than 70 nor more than 75 ppm. 1/
9.	Calcium disodium EDTA - not less than 70 nor more than 75 ppm. 1/
10.	Milk solids-not-fat 2/ - not less than 8.5 percent of the percent of moisture in the finished product.
11.	The oil separated from the finished product shall possess the following solid fat index values:

Solid fat index	Temperature °F	Temperature °C
28.0 - 32.0	50	10
18.0 - 21.0	70	21
17.5 - 20.5	80	27
11.5 - 13.0	92	33
7.5 - 9.5	100	38
5.5 - 8.0	105	41

1/ These tests shall be performed only when specified (see 6.2).

2/ Determine as follows:

$$\text{Milk solids-not-fat} = \frac{\text{Percent residue - percent salt x 100}}{\text{Percent moisture}}$$

3.4 Temperature of holding, transportation and delivery. When specified (see 6.2), the finished product shall be held below 45°F (7°C) until delivered.

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* 3.5 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to provisions of the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

3.6 Plant qualification. The product shall originate, be processed, marked and handled in plants which are operated under the requirements of MIL-STD-668.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Plant qualification conditions. The product furnished under this document shall be unacceptable if not produced and stored in plants which currently meet the qualification conditions of 3.6.

4.3 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection. In accordance with 4.1 above, components and material shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.1.1 Examination of cotton seed oil, salt, milk solids-non-fat and emulsifying agents. Determination of compliance of the above ingredients with requirements of table I, as concerns identity, shall be made by examination of labels, invoices, grade and salmonellae negative certificates or other valid documents. In addition, salt shall be examined for the condition requirement as concerns free flowing. The sample for examination shall be a 1 pound (0.45 kg) composite derived from five primary containers or all containers if less than five form a lot. Nonconformance to the free-flowing characteristic or identity requirements shall be cause for rejection of the lot or finished product made therefrom, as applicable.

4.3.1.2 Examination of water. Water shall be examined to determine compliance with the requirement of item 2, of table I as concerns approval source, by examination of state and local health certificates. In addition, water shall

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be examined organoleptically to determine compliance with condition requirement. Nonconformance to an approval source or condition requirement shall be cause for rejection of the lot or finished product made therefrom, as applicable.

4.3.2 Testing of components. Testing of components described in this specification shall be as shown in table IV. Results shall be applicable to the lot average. Nonconformance to one or more test requirements shall be cause for rejection of the lot.

TABLE IV. Testing of components

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported to	Test ref. and rqmt. par.
Cottonseed oil (see footnote 1/ to table I)	1/	Primary containers	1/	Peroxide value	Nearest 0.1	4.5.1 and 3.1
				Fat stability (A.O.M.)	See footnote 1/ to table I	

1/ The sample for testing shall be an approximate 1 pound (454 gram) composite derived from five primary containers or all containers if less than five form a lot. All samples shall be labeled clearly with adhesive labels bearing the name of the margarine manufacturer, contract number, time and date of sampling, lot number of oil and source of oil. Samples shall be shipped to the laboratory in dry ice.

4.3.3 Examination of end item. Examination shall be in accordance with tables VI and VII and 4.3.3.1 through 4.3.3.3. The sampling plans and acceptance criteria shall be as shown in table V. Acceptable quality levels (AQLs) shall be expressed in terms of defects per hundred units.

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TABLE V. Sampling plans and AQLs

Table	Inspection level	Sample unit	Lot size expressed in	AQLs Major	AQLs Minor
VI	S-3	One primary container	Primary containers	-	2.5
VII	S-2	One pound (0.45 kg) from a primary container	Primary containers	1.5	-

TABLE VI. Examination for net weight 1/

Category	Defect
Minor	
201	Net weight less than 6 pounds and 2 ounces (2.7 k).

1/ Report results to nearest 1/8 ounce (gram). Lot shall be rejected if sample data indicate lot average net weight less than 6 pounds, 4 ounces (2.8 kg).

TABLE VII. Examination of finished product

Category	Defect
Major	
151	Color not uniform yellow characteristic of colored commercial margarine.
152	Foreign flavor, such as rancid, tallowy or oxidized.
153	Not firm compact body of good spreadability at 60° to 70°F (16 to 21°C).
154	Crumbly, grainy, mealy, oily or visually moist.

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4.3.3.1 Examination of cans. Examination of cans for level A and C shall be in accordance with the applicable examination criteria contained in PPP-C-29.

4.3.3.2 Examination of can labeling. Examination of can for labeling shall be in accordance with the examination criteria contained in MIL-L-1497.

4.3.3.3 Examination for packing and shipping container. Examination of the filled and sealed shipping container shall be in accordance with the examination criteria contained or referenced in PPP-C-29.

4.4 Sampling procedure and acceptance criteria for testing of finished product. The finished product shall be tested to determine conformance to table II as concerns cube test and for analytical characteristics as specified in table III, in accordance with procedures referenced in 4.5. Lot size shall be expressed in terms of cans. The sample unit shall be one filled can that is sealed and the sample size shall be the number of units indicated by inspection level S-1. The cube test shall be determined initially on the individual sample units and the analytical characteristics shall be performed on a composite sample derived from the sample units used to perform the cube test. Each result shall be reported to the same decimal place or unit as specified in the pertinent requirements. Vitamin A shall be reported to the nearest 10 International Units and Vitamin D to the nearest fifty International Units. One or more test failures for analytical characteristics or cube test failure shall be cause for rejection of the lot.

4.5 Tests. Test procedures and controls which differ from those specified herein, may be used by the contractor if they provide a quality assurance equivalent to those specified. If the Government contracting officer determines that such procedures and controls do not provide, as a minimum, such quality assurance, the contractor will use the test procedures set forth herein. In case of dispute as to test results, the test methods specified herein will govern.

* 4.5.1 Chemical analyses. Chemical analyses shall be made in accordance with the methods from the publications listed below:

Test	Source	Method
Fat	Chapter: Dairy Products Section: Butter 1/	Official Final Action
Salt	Chapter: Dairy Products Section: Butter 1/	Official Final Action
Moisture	Chapter: Dairy Products Section: Butter	Official Final Action

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Test	Source		Method
Vitamin A	Chapter:	Vitamins and other nutrients 1/	Vitamin A in margarine
Vitamin D	Chapter:	Vitamins and other nutrients 1/	Vitamin D in margarine
Peroxide value	Chapter:	Commercial Fats and Oils 2/	Cd 8-53
	Section:	C	
Fat stability (active oxygen method)	Chapter:	Commercial Fats and Oils 2/	Cd 12-57
	Section:	C	
Propyl gallate	Chapter:	Food Additives: Direct	Official Final Action
	Section:	Antioxidants 1/	
Butylated hydroxyanisole	Chapter:	Food Additives: Direct	BHA
	Section:	Antioxidants 1/	
Butylated hydroxytoluene	Chapter:	Food Additives: Direct	BHT
	Section:	Antioxidants 1/	
Solid fat index	Chapter:	Commercial Fats and Oils 2/	Cd 10-57
	Section:	C	

1/ Official Methods of Analysis of the Association of Official Analytical Chemists.

2/ Official and Tentative Methods of the American Oil Chemists' Society.

4.5.2 Calcium disodium EDTA test. The procedure for determination of calcium disodium EDTA shall be as follows:

METHOD OF ANALYSIS FOR FOOD ADDITIVE DISODIUM CALCIUM EDTA

1. Scope:

This method is applicable to the determination of calcium EDTA (the calcium chelate of the disodium salt of ethylenediaminetetraacetic acid) in a variety of food products.

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2. Principle:

The sample is clarified by filtering after treating with barium carbonate and filter aid. The filtrate is treated with magnesium to remove interferences. Calcium EDTA is then determined in a filtered portion of the sample. In a strongly acidic solution, calcium EDTA reacts with zirconium, chelating an amount of zirconium equivalent to the calcium EDTA present. The excess zirconium is determined by its reaction with zylenol orange producing a red colored complex. The decrease in absorbance at 535 millimicrons is proportional to the calcium EDTA present.

3. Reagents:

(a) Filter acid. Celite, J-M- analytical filter aid.
(b) Barium carbonate. Reagent grade.
(c) Hydrochloric acid, 5N. Dilute 417 ml of concentrated hydrochloric acid to one liter.
(d) Magnesium chloride solution. Dissolve 200 grams of reagent grade magnesium chloride ($MgCl_2.6H_2O$) in 800 ml of distilled water.
(e) Ammonium hydroxide, concentrated reagent.
(f) Phosphoric acid, dilute. Dilute 10 ml of 85% reagent phosphoric acid to one liter.
(g) Hydroxylamine hydrochloride, approximately 10%. Dissolve 100 grams of hydroxylamine hydrochloride in water and dilute to one liter.
(h) Calcium EDTA standard solution A. Dissolve 1.101 grams of the calcium EDTA di-hydrate in water and dilute to one liter. One ml is equivalent to one mg of calcium EDTA.

Determination of calcium EDTA

Calcium EDTA standard solution B. Dilute 25.0 ml of solution A (3h) to volume in a 500-ml volumetric flask.

Zirconium standard solution A. Dissolve 3.454 grams of zirconium oxychloride ($ZrOCl_2.8H_2O$), in 100 ml of concentrated hydrochloric acid and dilute to one liter. 1/ Standardize the solution by the hydroxide procedure. One ml is approximately equivalent to one mg of zirconium.

Zirconium standard solution B. Dilute a volume of the zirconium standard A equal to 6 mg of zirconium and 20 ml of concentrated hydrochloric acid to one liter. Prepare fresh daily.

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Xylenol orange reagent. Dissolve 1.00 gram of xylenol orange in anhydrous methanol and dilute to one liter with methanol.

1/ Zirconium oxychloride and xylenol orange are available from Fisher Scientific Co. catalog Nl. X-11.

4. Apparatus:

Beakers, 250-ml.
Buchner Funnel, 9 cm.
Filtering apparatus, bell-jar type.
Steam bath.
Hot plate.
Volumetric flasks, 50-ml and 250-ml.
Pipets, various sizes.
Spectrophotometer, Beckman Model B or equivalent.
Absorption cells, Corex, 4.00-cm light path.
Filter paper, 9 cm Whatman No. 3 or equivalent.
Filtering crucible, 30-ml, fine porosity.

5. Preparation of standard curve:

a. Add 4.0 ml of 5N hydrochloric acid, 5.0 ml of 10% hydroxylamine hydrochloride, 5.0 ml of zirconium standard solution B and 4.0 ml of xylenol orange reagent to each of six 50-ml volumetric flasks. Prepare a color blank containing all reagents except the zirconium. Dilute 34 ml of the calcium EDTA standard solution B to 500 ml with distilled water. Pipet into the previously prepared volumetric flasks 0, 5, 10, 15, 20, and 25 ml of the freshly prepared standard (equivalent to 0, 17, 34, 51, 68, and 85 micrograms of calcium EDTA), dilute to volume and mix well.

b. Allow the standards to stand for 15 minutes for full color development. Zero the spectrophotometer on the color blank and read the absorbances of the standards at 535 millimicrons and a cell length of 4.0 cm.

c. Plot the absorbance against the micrograms of calcium EDTA on ordinary coordinate paper.

6. Preparation of samples:

To insure homogeneity of samples, each sample of product under test should be transferred completely to a Waring blender and blended until no visible evidence of segregation exists. With products consisting of an oil and water mixture, sampling should be performed while the sample is still being stirred in the blending step. Exercise caution so that the pipet does not contact the rotating blades. After blending, unknowns are determined as given below in section 7 (sample recovery curve). Necessary corrections for a given food product blank are made (see section 10).

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7. Sample recovery curve:

(a) Weigh 10.00 grams of sample into a 250-ml beaker, add 0, 5, 10, 12, 15, and 17 ml of calcium EDTA standard solution B from a buret. Dilute to 100 ml with distilled water.

(b) Add to each beaker 4.0 grams of filter aid and 2.0 grams of barium carbonate. Mix with a stirring rod and heat almost to boiling on a hot plate with continuous stirring.

(c) Transfer the samples to the steam bath and digest for one-half hour with stirring at 10 minute intervals.

(d) Remove the beakers from the steam bath, cool to room temperature in a cold water bath and filter on a Buchner funnel catching the filtrate in a clean beaker. (The filter should be prepared using a 9.0 cm Whatman No. 3 or equivalent paper and 2.0 grams of filter aid added as a slurry and pulled down under full vacuum). Wash the contents of the funnel by adding four successive 25-ml portions of water to the original beaker, policing down the sides with the first wash and pouring into the funnel.

(e) To the filtrate from 7d add 3.0 ml of 5N hydrochloric acid and 10 ml of magnesium chloride solution (3d). Neutralize the resulting solution to a pH of 9.0 with concentrated ammonium hydroxide and add 0.5 ml in excess. Add 5.0 ml of phosphoric acid (3f) to the solution while stirring vigorously.

(f) Transfer the filtrate to a 250-ml volumetric flask, rinse the beaker with distilled water, transfer into the flask, make to volume and mix thoroughly.

(g) Filter 75-100 ml of the solutions, after allowing them to stand for at least one hour, through a dry fine porosity filtering crucible or a dry 42 Whatman filter paper. Discard the first 25 ml of filtrate collecting the remaining filtrate in a clean dry beaker.

(h) Prepare a 50-ml volumetric flask as in (5a) for each sample. Also prepare another 50-ml volumetric flask for each sample to be used as a color blank (do not add zirconium to the sample-color reagent blank). Pipet a 25.0 ml aliquot into the second prepared volumetric flasks to be used as a sample color reagent blank. Dilute each flask to volume and mix well. Proceed with the determination as in (5c). Determine the absorbance of each sample using its color blank as the reference standard. Read the micrograms of calcium EDTA from the standard curve.

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Calculation of results:

$$\text{ppm calcium EDTA} = \frac{\text{micrograms of calcium EDTA found}}{\text{grams of sample x aliquot}}$$

(NOTE: If small amounts of oil are present after the filtration and washing step, they may be ignored since they will be removed in the second filtration. Any slight turbidity (encountered after the filtration steps) is ignored since each sample is compensated for by its own reagent color blank.)

10. Recovery data:

The food products listed below have been analyzed by this method with the given recoveries obtained: (Reference: Food Additive Regulation, Section 121.1017 and .1056, Food and Drug Administration.)

Product	No. of brands tested	Average blank - (ppm)	Average percent recovery (corrected)
Margarine	3	4	100
French dressing	3	6	95
Mayonnaise	3	1	94
Salad dressing	3	1	91

4.5.3 Cube test. Cube test shall be performed on a solid cube of margarine which has not been worked or molded. Procedure shall be as specified in 4.5.3.1 and 4.5.3.2.

4.5.3.1 Preparation of sample. Temper an undisturbed can of product at a temperature of 70°F to 75°F (16 to 24°C) for a sufficient length of time (24) hours to condition the margarine so that a test specimen can be obtained. Remove the margarine from the can by cutting out both ends of the can and sliding the can from the margarine. Cut the margarine into longitudinal sections approximately 2-1/2 inches (64 mm) thick and remove a 1 inch (25 mm) cube of product for testing.

4.5.3.2 Determination. Place the cube in a shallow, flat-bottomed, petri dish and cover cube with an inverted 250 ml beaker. Hold sample in a 90° ± 2°F (32°C ± 1°C) incubator for period of 40 hours. After this treatment there shall be no evidence of change in original dimensions of the cube or of free oil leakage.

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4.5.4 Potassium sorbate test. The procedure for determination of potassium sorbate shall be as follows:

ULTRAVIOLET METHOD OF ANALYSIS FOR THE DETERMINATION OF SORBIC ACID OR POTASSIUM SORBATE (OR BOTH) IN MARGARINE

This method is written for margarine containing from 0.05 to 0.20 percent by weight sorbic acid/potassium sorbate.

1. Reagents:

Standard sorbic acid solution - Dissolve 10 mg of sorbic acid in isopropanol in a 100 ml volumetric flask and dilute to volume.

Isopropanol, spectral grade.

2. Calibration:

Introduce a weighed sample (20 grams) of control unsupplemented margarine into a Waring blender. Add 1 ml of N hydrochloric acid and 75 ml of isopropanol. Blend for about 30 seconds at high speed (until sample is homogeneously dispersed).

Filter the mixture through Whatman No. 41 filter paper and collect about 30 ml of filtrate.

Transfer 1 ml of filtrate to each of four 100-ml volumetric flasks. Add 0, 1.0, 2.0 or 3.0 ml of the sorbic acid solution to each flask. These standards will correspond, respectively, to 0, 0.0010, 0.0020 and 0.0030 grams per liter of sorbic acid in the final dilution. Dilute to volume with isopropanol.

Obtain the absorbance of the solutions at 255 millimicrons in a 1-cm cell using isopropanol as the blank. Plot a graph of concentration of sorbic acid in grams per liter in the final dilution versus absorbance.

3. Sample determination:

The sample of margarine to be analyzed is treated as above except that no sorbic acid is added to the volumetric flasks. Refer absorbance for sample to standard curve for conversion to sorbic acid in grams per liter in the final dilution.

The sample data should be corrected by using a recovery factor. Percent recovery is obtained by adding a known amount of sorbic acid to the margarine and analyzing the sample as above.

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4. Calculations:

Sorbic acid (percent by weight in margarine) = C x F x 50

C = concentration of sorbic acid in grams per liter in final dilution from calibration curve.

F = recovery factor = $\frac{\text{grams theoretical of preservative}}{\text{grams found by analysis}}$

For potassium sorbate, multiply the sorbic acid value by 1.34.

5. PACKAGING

5.1 Preservation-packaging. A net weight of 6 pounds and 4 ounces (2.8 kg) of the product shall be packaged into a size 603 by 700 can, in accordance with the level A or C requirements of PPP-C-29, as specified (see 6.2).

5.2 Packing. The cans of the product shall be packed in accordance with levels A, B or C requirements of PPP-C-29, as specified (see 6.2).

5.3 Marking.

5.3.1 Cans. Cans shall be labeled in accordance with MIL-L-1497 and as follows:

MARGARINE (in letters larger than any other used on the label).

Ingredient statement
Antioxidant (by name) added
Artificial coloring (if used)
Artificial flavoring (if used)
15,000 I.U. (minimum) Vitamin A per pound (0.45 kg) added
1500 I.U. (minimum) Vitamin D per pound added (when applicable)
Net weight
Manufacturer's name and address
Date of pack

5.3.2 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129 and as specified in 5.3.2.1.

5.3.2.1 Additional marking. The following information shall be printed or stenciled on top of the shipping container in letters 1 to 1-1/2 inches (25 mm to 36 mm) high, all letters being of the same height:

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WHEN POSSIBLE STORE BELOW 45°F (7°C)

6. NOTES

6.1 Technical information. The product is designed to withstand a temperature of 95°F (35°C) with slight, if any, oil separation, which makes it possible to use the product for field use or emergency mass feeding. Because of the military necessity for a higher melting point than is built into regular consumer margarine, this product will have a more waxy consistency in the mouth. However, when product is eaten on bread, melted on hot food, or used in cooking, this greater waxiness should not detract from its usefulness. When refrigeration facilities are available in the military supply line, it is strongly recommended that the product be held below 45°F (7°C) for the optimum retention of flavor.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When Vitamin D is required (see tables I and III).
(c) When tests specified in items 5 through 9 of table III are required.
(d) When finished product must be held below 45°F (7°C) until delivered (see 3.5).
(e) Levels of packaging and packing required (see 5.1 and 5.2).

6.3 Changes from previous issue. The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 45

Review activities:

Army - MD
Navy - MC
DP-SS

Preparing activity:

Army - GL

Project No. 8945-0050

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MARGARINE, CANNED - MIL-M-10958F

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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